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324353 (25)

BE (3rd Semester)

Examination, Nov.-Dec., 2014

Branch : Electrical Engineering

BASIC ELECTRONICS (NEW)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) of each question is compulsory.

(ii) Solve any two parts from (b), (c), (d) in each question.

Q. 1. (a) Define drift current & diffusion current. 2

(b) Explain the behaviour of 'pn' junction diode under forward and reverse biasing condition. Give suitable diagrams wherever necessary. 7

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(c) A silicon diode conducts 5 mA at room temperature in forward biased condition at 0.7 volt. Calculate its reverse saturation current. If now forward voltage is increased to 0.75 volt, calculate the new current through the diode. 3+4

(d) Draw & explain VI characteristics of a forward biased and reverse biased pn junction diode. Also draw the VI characteristics of typical Ge & Si diode. 5+2

Q. 2. (a) Define avalanche breakdown and zener breakdown. 1+1

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(b) Explain the working of centre tapped transformer full wave rectifier. Draw the circuit diagram & voltages and load current waveforms. 7

(c) Draw the VI characteristics of zener diode for forward and reverse biased condition. Explain the application of zener diode as voltage regulator. 3+4

(d) A zener diode has V_z = 12 V, determine the minimum and maximum zener current as well as the output voltage : 3+3+1

(i) When the zener diode is considered to be an ideal one.

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(ii) When the zener resistance of zener diode is 7 ohms.

Comment on the results.

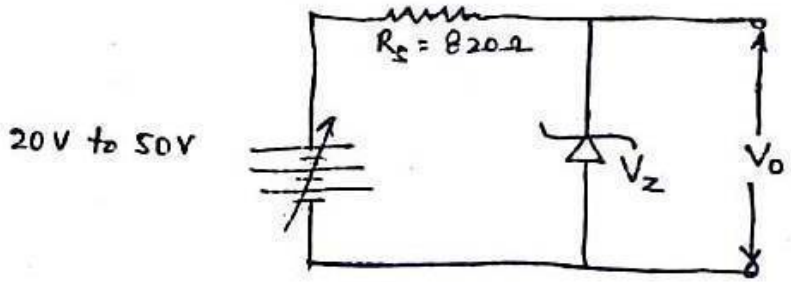

Fig. (1)

Q. 3. (a) Give the condition at Emitter junction and collector junction at Active, Saturation, Cut off and Inverted region of operation. 2

(b) Draw and explain the circuit, input and output characteristics of a transistor in (CB) common base configuration. 7

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(c) Explain the working of 'pnp' & 'npn' transistor. Draw the symbol of 'pnp' & 'npn' transistor. **6+1**

(d) Define α & β. Deduce the relationship between α & β. Find the value of β if : **2+2+3**

(i) $\alpha = 0.9$

(ii) $\alpha = 0.94$

(iii) $\alpha = 0.98$

Q. 4. (a) Define transconductance. **2**

(b) Explain the working and characteristics of depletion type MOSFET. **7**

(c) State the differences between FET and MOSFET. **7**

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(d) Prove that for JFET : **7**

$$g_m = \frac{-2I_{DSS}}{V_P}\left(1 - \frac{V_{GS}}{V_P}\right)$$

Q. 5. (a) What do you mean by positive feedback amplifier. **2**

(b) Define : **7**

(i) Series positive clipper

(ii) Shunt positive clipper

(iii) Series negative clipper

(iv) Shunt negative clipper

(c) Draw and explain the circuit and operation of LC Hartley oscillator. **7**

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(d) If L = 800 mH, C = 0.01 pf, R = 1000 ohm and C_m = 20 pf are the various values of an ac equivalent circuit of a piezo electric crystal. Determine series resonant frequency (f_s) and parallel resonant frequency (f_p) of the crystal. 7

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